

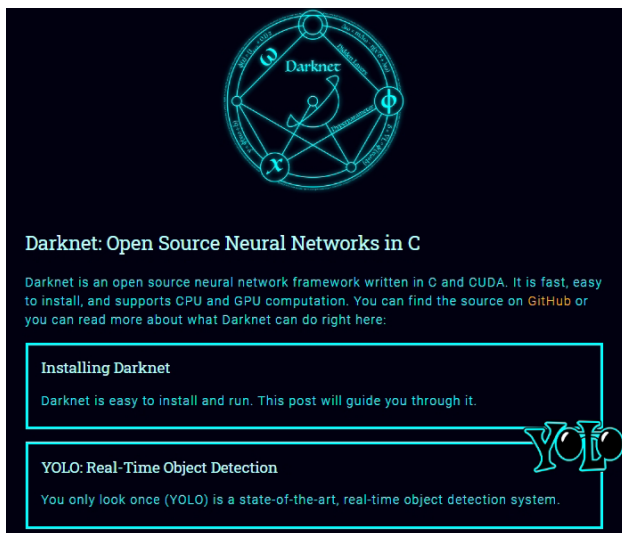


Figure 1: Home page of the Darknet platform

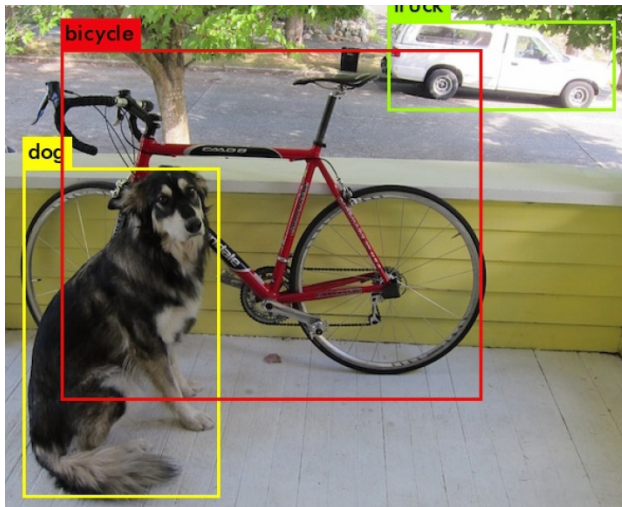


Figure 2: Real-time detection of objects

Darknet: An open source platform for neural networks in C

Darknet is a high performance open source framework for the implementation of neural networks. Written in C and CUDA, it can be integrated with CPUs and GPUs.

Advanced implementations of deep neural networks can be done using Darknet. These implementations include You Only Look Once (YOLO) for real-time object detection, ImageNet classification, recurrent neural networks (RNNs), and many others.

Installation and working with Darknet

Darknet can be installed directly with easy to use instructions. It needs two dependencies, both of which are optional based on the implementation scenario.

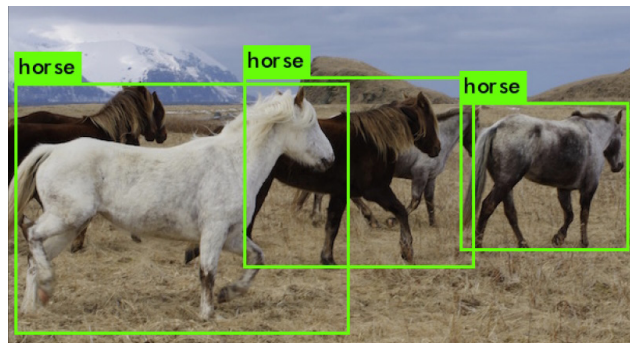


Figure 3: Output logs

Dependency 1: OpenCV for multiple types of images

Dependency 2: CUDA for GPU based computation

The base environment of Darknet can be installed using the following instructions:

Clone: `git clone https://github.com/pjreddie/darknet.git`

Move to Directory: `cd darknet`

make Instruction: `make`

Once all the instructions are executed successfully, you can run the environment using the following command:

```
./darknet
```

Real-time object detection using a pre-trained model

YOLO is one of the powerful methods of real-time object detection with integration of advanced deep learning. It makes use of convolutional neural networks (CNN) for the prediction of objects by using advanced mathematical formulations of image processing.

To work with real-time object detection, the data set in which the weights of pre-trained models are available is imported first. This is done so that the images of all real world objects can be mapped with the implementation for prediction. The weights of pre-trained models are downloaded as follows:

```
wget https://pjreddie.com/media/files/yolov3.weights
```

Next, the detection is done, specifying the image to be identified:

```
./darknet detect cfg/yolov3.cfg yolov3.weights data/dog.jpg
```

The output is generated with the dynamic fetching of the objects, with a label that marks their actual identity (Figures 2 and 3).

Continued On Page 73....